The module of derivations of graded Tjurina algebras

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1 Lecture 1- March 25, 2026

The Tjurina Algebra

Let

$$S = \mathbb{C}[x_1, \ldots, x_n].$$

For a polynomial $f \in S$, the *Jacobian ideal* of f is

$$J_f = \left\langle \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right\rangle.$$

Definition 1.1 (Tjurina algebra)**.** The *Tjurina algebra* of f is

$$T(f) = \frac{\mathbb{C}[x_1, \ldots, x_n]}{\langle f, J_f \rangle}.$$

The number

$$\tau(f) := \dim_{\mathbb{C}} T(f)$$

is called the *Tjurina number.*

Theorem 1.2 (Mather–Yau)**.** *Let $f, g \in \mathbb{C}\{x_1, \ldots, x_n\}$ define isolated hypersurface singularity germs at the origin. Then the germs*

$$(V(f), 0) \quad \textit{and} \quad (V(g), 0)$$

are analytically isomorphic if and only if their local Tjurina algebras

$$\frac{\mathbb{C}\{x_1, \ldots, x_n\}}{\langle f, J_f \rangle} \quad \textit{and} \quad \frac{\mathbb{C}\{x_1, \ldots, x_n\}}{\langle g, J_g \rangle}$$

are isomorphic as local $\mathbb{C}$-algebras.

Remark 1.3. The Mather–Yau theorem says that the analytic isomorphism type of an isolated hypersurface singularity is completely determined by its Tjurina algebra. Thus the finite-dimensional algebra $T(f)$ packages the local analytic information of the singularity.

Yau's Lie Algebra

Definition 1.4 (Yau algebra)**.** Let $f \in \mathbb{C}[x_1, \ldots, x_n]$ define an isolated hypersurface singularity. The *Yau algebra* associated to f is the Lie algebra

$$L(f) := \mathrm{Der}_{\mathbb{C}}(T(f)).$$

Theorem 1.5 (Yau, 1981)**.** *Let f define an isolated hypersurface singularity. Then*

$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f))$$

is a finite-dimensional solvable Lie algebra.

Remark 1.6. The phrase in the handwritten notes that appears as "$T(f)$ defines an isolated singularity which is solvable" is interpreted as the standard statement that the derivation Lie algebra of the Tjurina algebra of an isolated hypersurface singularity is solvable.

Program of Questions

The lecture notes outline the following sequence of questions.

(1) Does the Lie algebra
$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f))$$
distinguish isolated hypersurface singularities?

(2) If $f \in \mathbb{C}[\underline{x}]$ is weighted homogeneous, then $T(f)$ is graded and hence
$$L(f) = \bigoplus_{j \in \mathbb{Z}} L_j$$
is a graded Lie algebra.

(3) Yau's conjectural direction concerns the negative-degree part of this graded Lie algebra. In the notation of the notes, this appears as
$$L_j \neq 0 \quad \text{for certain } j < 0.$$

(4) If f is weighted homogeneous, then by Milnor–Orlik the dimension of the Tjurina algebra depends only on the weights:
$$\dim_{\mathbb{C}} T(f) \quad \text{depends only on the weights of } f.$$

(5) Yau–Zuo type conjectures concern bounds for
$$\dim_{\mathbb{C}} \mathrm{Der}_{\mathbb{C}}(L(f)).$$

(6) Higher-order versions of these invariants and questions can be formulated.

Complex Hypersurfaces

Basic Definitions

Let
$$f \in \mathbb{C}[x_1, \ldots, x_n].$$
The complex hypersurface defined by f is
$$V(f) = \{p \in \mathbb{C}^n \mid f(p) = 0\} \subsetneq \mathbb{C}^n.$$

Definition 1.7 (Singular locus)**.** The singular locus of the hypersurface $V(f)$ is
$$\mathrm{Sing}(f) = \left\{ p \in \mathbb{C}^n \;\middle|\; f(p) = 0 \text{ and } \frac{\partial f}{\partial x_i}(p) = 0 \text{ for all } i = 1, \ldots, n \right\}.$$
Equivalently,
$$\mathrm{Sing}(f) = V(f, J_f).$$

Definition 1.8 (Isolated hypersurface singularity)**.** The hypersurface $V(f)$ has an *isolated hypersurface singularity* at the origin if
$$\mathrm{Sing}(f) = \{\underline{0}\}$$
in a sufficiently small analytic neighborhood of the origin.

Examples

Example 1.9 (The cusp)**.** The plane curve

$$V(x^3 - y^2) \subseteq \mathbb{C}^2$$

has a cusp singularity at the origin.

Example 1.10 (The quadratic cone)**.** The hypersurface

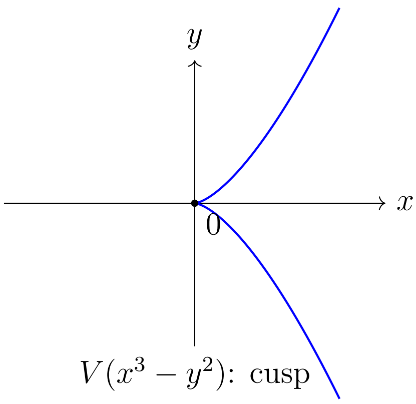
$$V(x^2 - y^2 - z^2) \subseteq \mathbb{C}^3$$

is a cone with an isolated singularity at the origin.

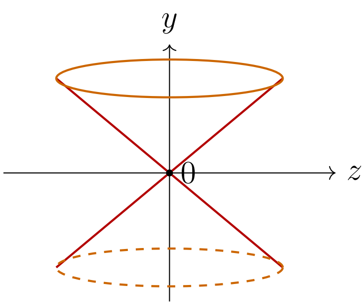
Example 1.11 (The Whitney umbrella)**.** The surface

$$V(x^2 z - y^2) \subseteq \mathbb{C}^3$$

is the Whitney umbrella.



$V(x^3 - y^2)$: cusp



$V(x^2 - y^2 - z^2)$: quadratic cone
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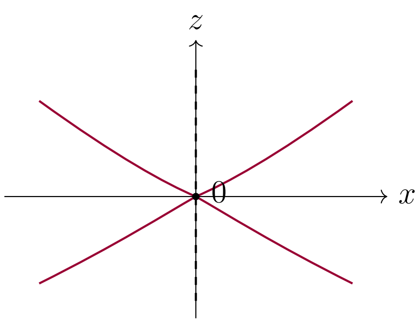

$V(x^2 z - y^2)$: Whitney umbrella
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Milnor and Tjurina Algebras

Definition 1.12 (Milnor algebra)**.** The *Milnor algebra* of f is

$$M(f) = \frac{\mathbb{C}[x_1, \ldots, x_n]}{J_f} = \frac{\mathbb{C}[x_1, \ldots, x_n]}{\left\langle \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right\rangle}.$$

Its complex vector space dimension

$$\mu(f) := \dim_{\mathbb{C}} M(f)$$

is called the *Milnor number*.

Definition 1.13 (Tjurina algebra)**.** The *Tjurina algebra* of f is

$$T(f) = \frac{\mathbb{C}[x_1, \ldots, x_n]}{\langle f, J_f \rangle} = \frac{\mathbb{C}[x_1, \ldots, x_n]}{\left\langle f, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right\rangle}$$

Its complex vector space dimension

$$\tau(f) := \dim_{\mathbb{C}} T(f)$$

is called the *Tjurina number*.

Remark 1.14. The Milnor number $\mu(f)$ measures the topology of the Milnor fiber. For an isolated hypersurface singularity, the Milnor fiber has the homotopy type of a bouquet of $\mu(f)$ spheres. The Tjurina number $\tau(f)$ measures the dimension of the base of the semi-universal deformation. If f is weighted homogeneous of weighted degree d with weights $w_1, \ldots, w_n$, and if f has an isolated singularity at the origin, then

$$\mu(f) = \tau(f).$$

Moreover, the Milnor number is determined by the weight system and the weighted degree. Equivalently, after normalizing the weights by d, it depends only on the normalized weights.

Derivations

Derivations of Algebras

Definition 1.15 (Derivation)**.** Let R be a k-algebra. A k-derivation of R is a k-linear map

$$D : R \to R$$

satisfying the Leibniz rule

$$D(ab) = aD(b) + bD(a)$$

for all $a, b \in R$. The set of all k-derivations of R is denoted by

$$\mathrm{Der}_k(R).$$

Example 1.16. Let
$$R = \mathbb{C}[x_1, \ldots, x_n].$$
Then each partial derivative
$$\partial_{x_i} : R \to R$$
is a $\mathbb{C}$-derivation. Moreover, every $\mathbb{C}$-derivation $D : R \to R$ can be written uniquely as
$$D = \sum_{i=1}^{n} D(x_i)\partial_{x_i}.$$
Therefore
$$\mathrm{Der}_{\mathbb{C}}(\mathbb{C}[x_1, \ldots, x_n]) \cong \bigoplus_{i=1}^{n} \mathbb{C}[x_1, \ldots, x_n]\partial_{x_i}$$
is a free R-module of rank n.

Derivations on Quotients

Question 1.17. How can one describe
$$\mathrm{Der}_{\mathbb{C}}\left(\frac{\mathbb{C}[x_1, \ldots, x_n]}{I}\right)?$$

A naive attempt is to descend the usual partial derivatives to the quotient. This does not always work.

Example 1.18. Let
$$A = \frac{\mathbb{C}[x]}{\langle x^2 \rangle}.$$
The usual derivative
$$\partial_x : \mathbb{C}[x] \to \mathbb{C}[x]$$
does not induce a well-defined derivation on A. Indeed,
$$[x^2] = [0] \in A,$$
but
$$\partial_x(x^2) = 2x,$$
and
$$[2x] \neq [0] \in A.$$
Equivalently,
$$\partial_x(\langle x^2 \rangle) \nsubseteq \langle x^2 \rangle.$$

Definition 1.19 (Derivations preserving an ideal)**.** Let $I \subseteq \mathbb{C}[x_1, \ldots, x_n]$. Define
$$\mathrm{Der}_I(\mathbb{C}[x_1, \ldots, x_n]) = \left\{ D \in \mathrm{Der}_{\mathbb{C}}(\mathbb{C}[x_1, \ldots, x_n]) \mid D(I) \subseteq I \right\}.$$
These are the derivations preserving the ideal I.

Proposition 1.20. *Let*
$$S = \mathbb{C}[x_1, \ldots, x_n]$$
and let $I \subseteq S$ be an ideal. Then there is a canonical isomorphism
$$\mathrm{Der}_{\mathbb{C}}\left(\frac{S}{I}\right) \cong \frac{\mathrm{Der}_I(S)}{I\,\mathrm{Der}_{\mathbb{C}}(S)}.$$

Proof. Every derivation $D \in \mathrm{Der}_I(S)$ induces a derivation

$$\overline{D} : S/I \to S/I$$

by

$$\overline{D}([f]) = [D(f)].$$

This is well-defined precisely because $D(I) \subseteq I$.

Thus we obtain a natural map

$$\mathrm{Der}_I(S) \longrightarrow \mathrm{Der}_{\mathbb{C}}(S/I).$$

Its kernel consists of those derivations D such that

$$D(S) \subseteq I.$$

Since

$$\mathrm{Der}_{\mathbb{C}}(S) = \bigoplus_{i=1}^{n} S\partial_{x_i},$$

this kernel is exactly

$$I\,\mathrm{Der}_{\mathbb{C}}(S).$$

Therefore the induced map gives the desired isomorphism

$$\frac{\mathrm{Der}_I(S)}{I\,\mathrm{Der}_{\mathbb{C}}(S)} \cong \mathrm{Der}_{\mathbb{C}}(S/I).$$

□

$$\begin{tikzcd}
S \arrow[r, "D"] \arrow[d, two heads] & S \arrow[d, two heads] \\
S/I \arrow[r, "\overline{D}"'] & S/I
\end{tikzcd}$$

Lie Algebras

Basic Definitions

Definition 1.21 (Lie algebra)**.** A Lie algebra over $\mathbb{C}$ is a $\mathbb{C}$-vector space L together with a bilinear map

$$[-,-] : L \times L \to L, \qquad (X,Y) \mapsto [X,Y],$$

called the *Lie bracket*, satisfying:

i) $[X,X] = 0$ for all $X \in L$.

ii) The Jacobi identity:

$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$$

for all $X, Y, Z \in L$.

Remark 1.22. Since the base field is $\mathbb{C}$, the condition $[X,X] = 0$ is equivalent to skew-symmetry:

$$[X,Y] = -[Y,X].$$

Example 1.23 (Derivations form a Lie algebra)**.** Let A be a $\mathbb{C}$-algebra. Then

$$\mathrm{Der}_{\mathbb{C}}(A)$$

is a Lie algebra with bracket

$$[D, E] = D \circ E - E \circ D.$$

Definition 1.24 (Homomorphism of Lie algebras)**.** Let L_1 and L_2 be Lie algebras. A map

$$\Phi : L_1 \to L_2$$

is a homomorphism of Lie algebras if:

i) Φ is $\mathbb{C}$-linear.

ii) For all $X, Y \in L_1$,
$$\Phi([X, Y]_{L_1}) = [\Phi(X), \Phi(Y)]_{L_2}.$$

If Φ is bijective, then Φ is an isomorphism of Lie algebras.

Derivation Lie Algebras of Tjurina Algebras

Let $f \in \mathbb{C}[x_1, \ldots, x_n]$ define an isolated hypersurface singularity. The central algebraic object in these notes is

$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f)).$$

Since

$$T(f) = \frac{S}{\langle f, J_f \rangle},$$

the quotient description of derivations gives

$$L(f) \cong \frac{\mathrm{Der}_{\langle f, J_f \rangle}(S)}{\langle f, J_f \rangle \, \mathrm{Der}_{\mathbb{C}}(S)}.$$

Remark 1.25. This formula allows one to compute $L(f)$ explicitly. A derivation

$$D = \sum_{i=1}^{n} a_i \partial_{x_i}$$

descends to $T(f)$ precisely when it preserves the Tjurina ideal:

$$D(\langle f, J_f \rangle) \subseteq \langle f, J_f \rangle.$$

Weighted Homogeneous Case

Definition 1.26 (Weighted homogeneous polynomial)**.** Let $w_1, \ldots, w_n$ be positive rational weights. A polynomial

$$f \in \mathbb{C}[x_1, \ldots, x_n]$$

is weighted homogeneous of weighted degree d if each monomial

$$x_1^{a_1} \cdots x_n^{a_n}$$

appearing in f satisfies

$$a_1 w_1 + \cdots + a_n w_n = d.$$

Remark 1.27. If f is weighted homogeneous, then the Euler derivation

$$E = \sum_{i=1}^{n} w_i x_i \partial_{x_i}$$

satisfies the Euler identity

$$E(f) = df.$$

Consequently,

$$f \in J_f$$

whenever the weighted degree d is nonzero in the base field. Over $\mathbb{C}$, this always holds. Thus, in the weighted homogeneous isolated hypersurface case,

$$T(f) = M(f).$$

After clearing denominators, we may assume the weights $w_1, \dots, w_n$ are positive integers. Then S, $T(f)$, and

$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f))$$

inherit $\mathbb{Z}$-gradings:

$$L(f) = \bigoplus_{j \in \mathbb{Z}} L_j.$$

Absolutely Isolated Surface Singularities

Double Points

Definition 1.28 (Double point)**.** Let

$$f = f_2 + f_3 + f_4 + \cdots$$

be the decomposition of a convergent power series into homogeneous parts, and assume $f(0) = 0$ and $df(0) = 0$. If

$$f_2 \neq 0,$$

then the hypersurface singularity $V(f)$ at the origin has multiplicity 2 and is called a double point.

Definition 1.29 (Absolutely isolated singularity)**.** Let

$$X = V(f) \subseteq \mathbb{C}^n$$

and suppose $0 \in X$ is an isolated singularity. The singularity is called *absolutely isolated* if a resolution of singularities of X can be obtained by blowing up only points.

Example 1.30. The surface

$$V(x^2 - y^2 - z^2)$$

is an absolutely isolated singularity.

Example 1.31 (The Whitney umbrella)**.** The surface

$$V(x^2 z - y^2) \subseteq \mathbb{C}^3$$

is the Whitney umbrella. Its singular locus is the line

$$\{x = 0,\ y = 0\}.$$

Thus it is not an isolated hypersurface singularity.

ADE Classification

Theorem 1.32 (ADE classification of rational double points)**.** *Let*

$$X = V(f) \subseteq \mathbb{C}^3$$

be a normal surface singularity at the origin. Then X is a rational double point, equivalently a Du Val singularity, if and only if it is analytically isomorphic to exactly one of the following hypersurface singularities:

$$A_k: \quad x^{k+1} + y^2 + z^2 = 0, \qquad k \geq 1,$$

$$D_k: \quad x^{k-1} + xy^2 + z^2 = 0, \qquad k \geq 4,$$

$$E_6: \quad x^3 + y^4 + z^2 = 0,$$

$$E_7: \quad x^3 + xy^3 + z^2 = 0,$$

$$E_8: \quad x^3 + y^5 + z^2 = 0.$$

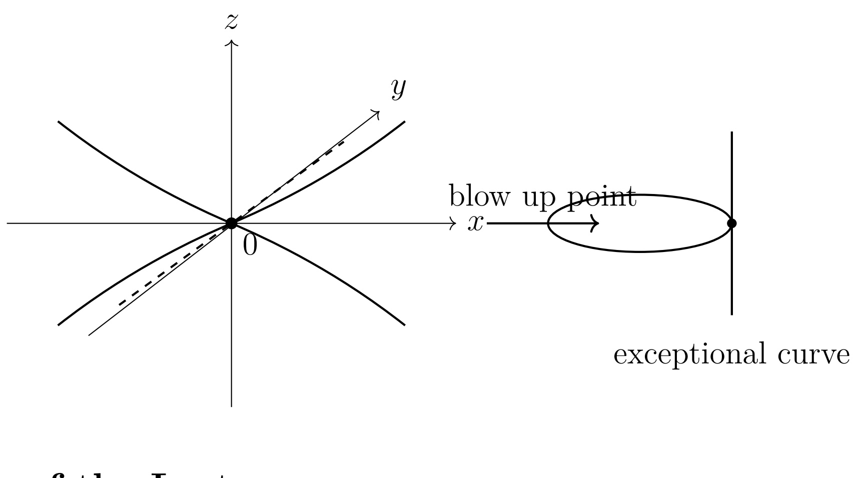
Remark 1.33. The ADE singularities are also called simple surface singularities or rational double points. Their classification is governed by the simply-laced Dynkin diagrams

$$A_k, \quad D_k, \quad E_6, \quad E_7, \quad E_8.$$

These diagrams also appear as the dual intersection graphs of exceptional divisors in the minimal resolution.

Schematic Picture

The following schematic diagram represents the idea that a double point surface singularity has a distinguished singular point at the origin, and resolution may proceed by blowing up points in the absolutely isolated case.



Summary of the Lecture

The notes establish the following chain of ideas.

(1) A hypersurface singularity

$$V(f) \subseteq \mathbb{C}^n$$

has an associated Jacobian ideal J_f.

(2) The Tjurina algebra

$$T(f) = \frac{\mathbb{C}[x_1, \ldots, x_n]}{\langle f, J_f \rangle}$$

is finite-dimensional when f defines an isolated hypersurface singularity.

(3) By the Mather–Yau theorem, the Tjurina algebra determines the analytic isomorphism class of the isolated hypersurface singularity.

(4) The derivation Lie algebra

$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f))$$

is a secondary invariant attached to the singularity.

(5) For weighted homogeneous singularities, $L(f)$ inherits a grading

$$L(f) = \bigoplus_j L_j.$$

(6) In surface singularity theory, absolutely isolated double points are classified by the ADE list:

$$A_k, \quad D_k, \quad E_6, \quad E_7, \quad E_8.$$

2 Lecture 2- March 26, 2026

Let $R = \mathbb{C}[x_1, \ldots, x_n]$ or $R = \mathbb{C}\{x_1, \ldots, x_n\}$. Let $f \in R$ and $X = \{f = 0\}$.

Definition 2.1 (Jacobian ideal)**.** The Jacobian ideal of f is

$$J_f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) \subseteq R.$$

Definition 2.2 (Tjurina algebra)**.** The Tjurina algebra of the hypersurface singularity X is

$$\mathcal{T}(X) = \mathcal{T}(f) = \frac{R}{(f, J_f)} = \frac{R}{\left(f, \frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)}.$$

Definition 2.3 (Yau algebra)**.** The Lie algebra associated to X, often called the Yau algebra in this context, is

$$\mathcal{L}(X) = \mathrm{Der}_{\mathbb{C}}(\mathcal{T}(X)).$$

Recall that a derivation is a linear map $D : \mathcal{T}(X) \longrightarrow \mathcal{T}(X)$ satisfying the Leibniz rule:

$$D(ab) = aD(b) + bD(a).$$

The Lie bracket is given by the commutator:

$$[D_1, D_2] = D_1 D_2 - D_2 D_1.$$

Remark 2.4 (Added Context)*.* If $\mathcal{T}(X) = R/I$, then a derivation D' can be lifted to a derivation D satisfying

$$D(I) \subseteq I.$$

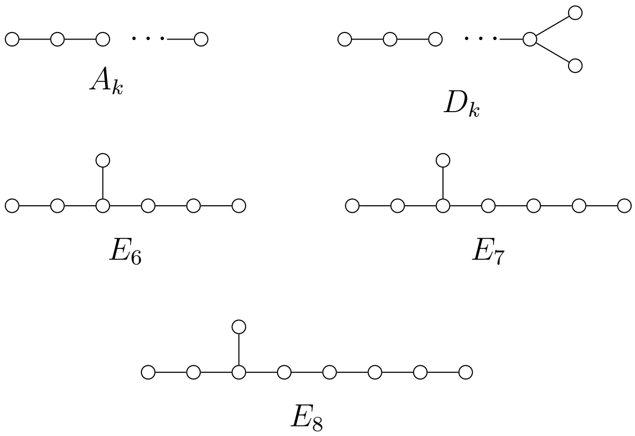
This descends to a derivation of $\mathcal{T}(X)$. This is the basic mechanism used below to compute $\mathcal{L}(A_k)$.

ADE Surface Singularities

The ADE surface singularities in $\mathbb{C}^3$ are the following hypersurfaces.

$$\begin{aligned} A_k: &\quad x^{k+1}+y^2+z^2=0, \qquad k\geq 1,\\ D_k: &\quad x^{k-1}+xy^2+z^2=0, \qquad k\geq 4,\\ E_6: &\quad x^3+y^4+z^2=0,\\ E_7: &\quad x^3+xy^3+z^2=0,\\ E_8: &\quad x^3+y^5+z^2=0. \end{aligned}$$

These are pairwise non-isomorphic hypersurface singularities.



Theorem 2.5 (Elashvili–Khimshiashvili)**.** *Let X and Y be ADE singularities. Except for the exceptional pair*

$$\{X,Y\}=\{A_6,D_5\},$$

we have

$$X\simeq Y \quad\Longleftrightarrow\quad \mathcal{L}(X)\simeq\mathcal{L}(Y)$$

Remark 2.6. The exceptional phenomenon is

$$A_6\not\simeq D_5, \qquad \mathcal{L}(A_6)\simeq\mathcal{L}(D_5).$$

This means the Yau algebra distinguishes almost all ADE singularities, but not quite.

Computation of $\mathcal{L}(A_k)$

Let

$$A_k=\{x^{k+1}+y^2+z^2=0\}.$$

We have $f=x^{k+1}+y^2+z^2$. The Jacobian ideal is

$$J_f=(x^k,y,z),$$

because

$$\frac{\partial f}{\partial x}=(k+1)x^k, \qquad \frac{\partial f}{\partial y}=2y, \qquad \frac{\partial f}{\partial z}=2z.$$

Hence the Tjurina algebra is

$$\mathcal{T}(A_k) = \frac{\mathbb{C}[x,y,z]}{(x^{k+1}+y^2+z^2, x^k, y, z)} \cong \frac{\mathbb{C}[x]}{(x^k)}.$$

A basis is given by $\{\overline{1}, \overline{x}, \ldots, \overline{x^{k-1}}\}$.

Proposition 2.7. *There is a* $\mathbb{C}$*-basis*

$$\mathcal{L}(A_k) = \operatorname{span}_{\mathbb{C}} \left\{ \overline{x\partial_x}, \overline{x^2\partial_x}, \ldots, \overline{x^{k-1}\partial_x} \right\}.$$

In particular, $\dim_{\mathbb{C}} \mathcal{L}(A_k) = k-1$.

Proof. Let

$$A = \frac{\mathbb{C}[x]}{(x^k)}.$$

A derivation $\overline{D} : A \to A$ lifts to $D' : \mathbb{C}[x] \to \mathbb{C}[x]$ such that

$$D'\big((x^k)\big) \subseteq (x^k).$$

We can write $D'(g) = \frac{\partial g}{\partial x} D'(x)$. Write

$$D'(x) = \lambda_0 + \lambda_1 x + \cdots + \lambda_{k-1} x^{k-1} \qquad \text{modulo } (x^k),$$

with the condition $D'(x^k) \in (x^k)$. We have

$$D'(x^k) = kx^{k-1} D'(x).$$

Modulo (x^k), this becomes

$$kx^{k-1} D'(x) \equiv k\lambda_0 x^{k-1}.$$

For this to be zero, we need $k\lambda_0 \overline{x}^{k-1} = 0$ in A. Thus, we get

$$\lambda_0 = 0.$$

So $D'(x) \in (x)/(x^k)$, which means the derivations are generated by

$$x\partial_x,\ x^2\partial_x,\ \ldots,\ x^{k-1}\partial_x.$$

These are linearly independent on A. Therefore they form a basis of $\mathcal{L}(A_k)$, and

$$\dim_{\mathbb{C}} \mathcal{L}(A_k) = k-1.$$

□

Definition 2.8. For $1 \le i \le k-1$, set

$$e_i = x^i \partial_x \in \mathcal{L}(A_k).$$

Then $\mathcal{L}(A_k) = \operatorname{span}_{\mathbb{C}}\{e_1, \ldots, e_{k-1}\}$.

Lemma 2.9. *For* $1 \le i, j \le k-1$*, the Lie bracket in* $\mathcal{L}(A_k)$ *is*

$$[e_i, e_j] = (j-i)e_{i+j-1},$$

with the convention $e_m = 0$ *if* $m \ge k$.

Proof. Using

$$e_i = x^i \partial_x, \qquad e_j = x^j \partial_x,$$

we compute

$$\begin{aligned} [x^i\partial_x, x^j\partial_x] &= x^i\partial_x(x^j)\partial_x - x^j\partial_x(x^i)\partial_x \\ &= jx^{i+j-1}\partial_x - ix^{i+j-1}\partial_x \\ &= (j-i)x^{i+j-1}\partial_x. \end{aligned}$$

Thus $[e_i, e_j] = (j-i)e_{i+j-1}$. □

Dimensions of the ADE Yau Algebras

The lecture records the following dimensions:

Singularity	$\dim_{\mathbb{C}} \mathcal{L}(\cdot)$
A_k	$k-1$
D_k	k
E_6	7
E_7	8
E_8	10

Equivalently,

$$\begin{aligned} &\dim_{\mathbb{C}} \mathcal{L}(A_k) = k-1, \quad \dim_{\mathbb{C}} \mathcal{L}(D_k) = k, \\ &\dim_{\mathbb{C}} \mathcal{L}(E_6) = 7, \qquad \dim_{\mathbb{C}} \mathcal{L}(E_7) = 8, \quad \dim_{\mathbb{C}} \mathcal{L}(E_8) = 10. \end{aligned}$$

These dimensions distinguish most ADE singularities by their associated Lie algebras. The dimension count alone leaves the following possible collisions:

$$\mathcal{L}(A_k) \quad \text{and} \quad \mathcal{L}(D_{k-1}) \qquad (k=5 \text{ or } k \geq 7),$$

and possible collisions with exceptional singularities:

$$(D_7, E_6), \quad (D_{10}, E_8), \quad (A_8, E_6), \quad (A_{11}, E_8),$$
$$(A_9, E_7), \qquad (D_8, E_7).$$

Remark 2.10. The notes indicate that the above equal-dimensional cases are distinguished by finer Lie-theoretic invariants, such as the upper central series, Cartan rank, indecomposability, and nilpotent ideals.

Nilpotent Lie Algebras and Cartan Subalgebras

Definition 2.11 (Lower central series)**.** Let L be a finite-dimensional Lie algebra. Define

$$L^0 = L, \qquad L^1 = [L, L],$$
$$L^n = [L, L^{n-1}] = \operatorname{span}_{\mathbb{C}}\{[x,y] \mid x \in L,\ y \in L^{n-1}\} \qquad (n \geq 1).$$

We say L is nilpotent if $L^n = 0$ for some n.

Remark 2.12. Some notes use the indexing

$$L_0 = L, \qquad L_1 = [L, L], \qquad L_2 = [L, L_1], \qquad \ldots.$$

Definition 2.13 (Cartan subalgebra)**.** Let L be a finite-dimensional Lie algebra. A Cartan subalgebra of L is a nilpotent subalgebra H satisfying the self-normalizing condition

$$N_L(H) = H,$$

where

$$N_L(H) = \{y \in L \mid [x,y] \in H \text{ for all } x \in H\}.$$

That is, if $[x,y] \in H$ for all $x \in H \implies y \in H$.

Theorem 2.14. *All Cartan subalgebras of a finite-dimensional complex Lie algebra have the same dimension. This common dimension is called the rank of L, denoted*

$$\operatorname{rk}(L).$$

Cartan Rank of $\mathcal{L}(A_k)$

Recall that

$$\mathcal{L}(A_k) = \mathrm{span}_{\mathbb{C}}\{e_1, \ldots, e_{k-1}\}, \qquad e_i = x^i \partial_x.$$

Let $H = \mathrm{span}_{\mathbb{C}}\{e_1\}$.

Claim 2.15. *The subspace H is a Cartan subalgebra of $\mathcal{L}(A_k)$. In particular,*

$$\mathrm{rk}(\mathcal{L}(A_k)) = 1.$$

Proof. First, H is nilpotent because

$$[H, H] = \mathrm{span}_{\mathbb{C}}\{[e_1, e_1]\} = 0.$$

Now we check the self-normalizing condition. Let

$$y = \sum_{i=1}^{k-1} \lambda_i e_i \in \mathcal{L}(A_k).$$

Since $[e_1, e_i] = (i-1)e_i$, we have

$$\begin{aligned} [e_1, y] &= \left[e_1, \sum_{i=1}^{k-1} \lambda_i e_i \right] \\ &= \sum_{i=1}^{k-1} \lambda_i [e_1, e_i] \\ &= \sum_{i=2}^{k-1} \lambda_i (i-1) e_i. \end{aligned}$$

If $[e_1, y] \in H = \mathrm{span}_{\mathbb{C}}\{e_1\}$, then the coefficients must vanish for $i \geq 2$. Since $i - 1 \neq 0$ for $i \geq 2$, we have

$$\lambda_i = 0 \qquad \text{for all } i \geq 2.$$

Thus $y = \lambda_1 e_1 \in H$. □

The lecture notes also record the analogous rank statements:

$$\mathrm{rk}(\mathcal{L}(D_k)) = 1,$$

$$\mathrm{rk}(\mathcal{L}(E_6)) = \mathrm{rk}(\mathcal{L}(E_8)) = 2.$$

Remark 2.16 (Added Context). The notes emphasize that Cartan rank helps distinguish some Lie algebras that have the same dimension. Dimension is a coarse invariant, while Cartan rank and central-series data are finer invariants.

The Exceptional Pair A_6 and D_5

The main exceptional case is

$$A_6 \not\simeq D_5, \qquad \mathcal{L}(A_6) \simeq \mathcal{L}(D_5).$$

For A_6, we have

$$\mathcal{T}(A_6) \cong \frac{\mathbb{C}[x]}{(x^6)}.$$

$$\mathcal{L}(A_6) = \mathrm{span}_{\mathbb{C}}\{e_1, e_2, e_3, e_4, e_5\},$$

where $e_i = x^i \partial_x$.

For D_5, the hypersurface is

$$D_5 = \{x^4 + xy^2 + z^2 = 0\}.$$

The derivations are spanned by:

$$\begin{aligned}
d_1 &= 3x\partial_x + 2y\partial_y, \\
d_2 &= 4y^2\partial_x + x\partial_y, \\
d_3 &= y^3\partial_x, \\
d_4 &= y^2\partial_y, \\
d_5 &= y^3\partial_y.
\end{aligned}$$

So

$$\mathcal{L}(D_5) = \mathrm{span}_{\mathbb{C}}\{d_1, d_2, d_3, d_4, d_5\}.$$

Theorem 2.17. *There is an isomorphism of Lie algebras*

$$\mathcal{L}(A_6) \cong \mathcal{L}(D_5).$$

Proof. Using the bases

$$\mathcal{L}(A_6) = \mathrm{span}_{\mathbb{C}}\{e_1, e_2, e_3, e_4, e_5\}$$

and

$$\mathcal{L}(D_5) = \mathrm{span}_{\mathbb{C}}\{d_1, d_2, d_3, d_4, d_5\},$$

define a linear map $\Phi : \mathcal{L}(A_6) \longrightarrow \mathcal{L}(D_5)$ by

$$\begin{aligned}
&e_1 \longmapsto d_1, \quad e_2 \longmapsto d_2, \qquad e_3 \longmapsto -\frac{1}{8}d_3, \\
&e_4 \longmapsto d_4, \quad e_5 \longmapsto -\frac{1}{2}d_5.
\end{aligned}$$

The lecture notes state that, after computing all commutators $[-,-]$ among the d_i, one has

$$\Phi([e_i, e_j]) = [\Phi(e_i), \Phi(e_j)] \qquad \text{for all } 1 \le i, j \le 5.$$

□

Remark 2.18. This proves the exceptional behavior:

$$A_6 \not\simeq D_5, \qquad \mathcal{L}(A_6) \simeq \mathcal{L}(D_5).$$

This is the single exceptional pair.

Summary of the Lecture

The lecture establishes the following chain of ideas.

(1) To a hypersurface singularity X, one attaches its Tjurina algebra

$$\mathcal{T}(X) = \frac{\mathbb{C}[x_1, \ldots, x_n]}{(f, J_f)}.$$

(2) The derivation Lie algebra

$$\mathcal{L}(X) = \mathrm{Der}_{\mathbb{C}}(\mathcal{T}(X))$$

is finite-dimensional.

(3) For the ADE singularities, $\mathcal{L}(X)$ almost determines X.

(4) The dimensions are

X	$\dim_{\mathbb{C}} \mathcal{L}(X)$
A_k	$k-1$
D_k	k
E_6	7
E_7	8
E_8	10

(5) When dimensions coincide, one uses finer invariants such as Cartan rank, central series, indecomposability, and nilpotent ideals.

(6) The only exceptional ADE pair is

$$A_6,\ D_5,$$

where

$$A_6 \not\simeq D_5 \quad \text{but} \quad \mathcal{L}(A_6) \simeq \mathcal{L}(D_5).$$